

Exercise 1

For each of the following formulas use the DPLL procedure to determine whether ϕ is satisfiable or unsatisfiable. Give a complete trace of the algorithm, showing the simplified formula for each recursive call of the DPLL function. Assume that DPLL selects variables in alphabetical order (i.e. $A, B, C, D, E, \dots$), and that the splitting rule first attempts the value False (F) and then the value True (T).

(a) $\phi_1 = (A \vee B) \wedge (B \vee C) \wedge (\neg B \vee D) \wedge (\neg A \vee \neg D) \wedge (\neg C \vee E) \wedge (A \vee \neg B \vee \neg D)$

(b) $\phi_2 = (\neg A \vee \neg B \vee \neg C) \wedge (A \vee \neg B) \wedge (A \vee \neg D) \wedge (B \vee \neg E) \wedge (\neg C \vee D) \wedge (C \vee E) \wedge (C \vee \neg E) \wedge (\neg D \vee E)$

Solution:

a) DPLL trace:

$$\{\{A, B\}, \{B, C\}, \{\neg B, D\}, \{\neg A, \neg D\}, \{\neg C, E\}, \{A, \neg B, \neg D\}\}$$

Pure literal E:

$$\{\{A, B\}, \{B, C\}, \{\neg B, D\}, \{\neg A, \neg D\}, \{A, \neg B, \neg D\}\}$$

Pure literal C:

$$\{\{A, B\}, \{\neg B, D\}, \{\neg A, \neg D\}, \{A, \neg B, \neg D\}\}$$

1. Splitting rule:

1a. $A \mapsto F$

$$\{\{B\}, \{\neg B, D\}, \neg B, \neg D\}$$

1b. $A \mapsto T$

$$\{\{\neg B, D\}, \{\neg D\}\}$$

2a. UP rule: $B \mapsto T$

$$\{\{D\}, \{\neg D\}\}$$

2b. UP rule: $D \mapsto F$

$$\{\{\neg B\}\}$$

3a. UP rule: $D \mapsto F$

$$\{\square\}$$

3b. UP rule: $B \mapsto F$

$$\{\}$$

Satisfying assignment: $A, \neg B, C, \neg D, E$

b) DPLL trace:

$$\{\{\neg A, \neg B, \neg C\}, \{A, \neg B\}, \{A, \neg D\}, \{B, \neg E\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

1. Splitting rule:

1a. $A \mapsto F$

$$\{\{\neg B\}, \{\neg D\}, \{B, \neg E\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

2a. UP rule: $B \mapsto F$

$$\{\{\neg D\}, \{\neg E\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

3a. UP rule: $D \mapsto F$

$$\{\{\neg E\}, \{\neg C\}, \{C, E\}, \{C, \neg E\}\}$$

4a. UP rule: $C \mapsto F$

$$\{\{\neg E\}, \{E\}, \{\neg E\}\}$$

5a. UP rule: $E \mapsto F$

$$\{\square\}$$

1. Splitting rule:

1b. $A \mapsto T$

$$\{\{\neg B, \neg C\}, \{B, \neg E\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

2b. Splitting rule:

2ba. $B \mapsto F$

$$\{\{\neg E\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

2bb. $B \mapsto T$

$$\{\{\neg C\}, \{\neg C, D\}, \{C, E\}, \{C, \neg E\}, \{\neg D, E\}\}$$

3ba. UP rule: $E \mapsto F$

$$\{\{\neg C, D\}, \{C\}, \{\neg D\}\}$$

3bb. UP rule: $C \mapsto F$

$$\{\{E\}, \{\neg E\}, \{\neg D, E\}\}$$

4ba. UP rule: $C \mapsto T$

$$\{\{D\}, \{\neg D\}\}$$

4bb. UP rule: $E \mapsto F$

$$\{\square, \{\neg D\}\}$$

5ba. UP rule: $D \mapsto F$

$$\{\square\}$$

5bb. UP rule: $D \mapsto F$

$$\{\square\}$$

There is no satisfying assignment.

Exercise 2

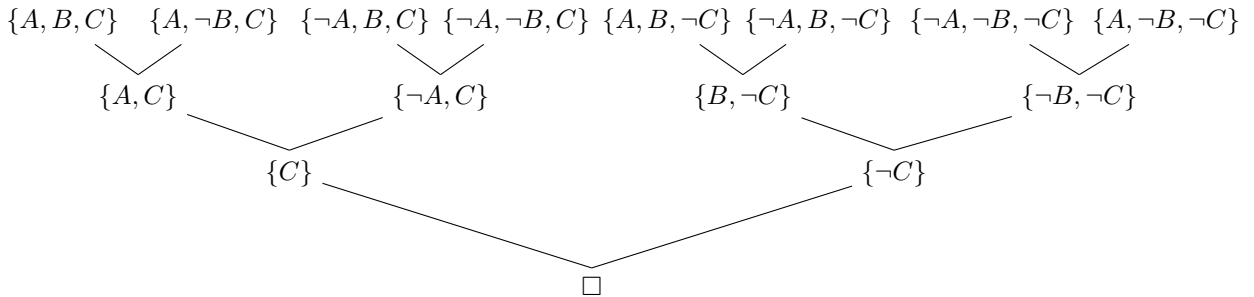
For the two formulas below, use resolution to prove that the formulas are unsatisfiable. To do so, first give a set of clauses Δ that is equivalent to the formula and second, use resolution to prove that it is unsatisfiable. Write the resolution process in the form of a tree for easier readability.

(a) $\phi_1 = (A \vee B \vee C) \wedge (\neg A \vee B \vee C) \wedge (A \vee \neg B \vee C) \wedge (A \vee B \vee \neg C) \wedge (\neg A \vee \neg B \vee C) \wedge (\neg A \vee B \vee \neg C) \wedge (A \vee \neg B \vee \neg C) \wedge (\neg A \vee \neg B \vee \neg C)$

(b) $\phi_2 = (G \vee \neg H) \wedge (\neg G \vee H) \wedge (A \rightarrow B) \wedge (\neg B \vee C) \wedge (A \vee D) \wedge (\neg C \vee \neg A \vee \neg E) \wedge (\neg D \vee A) \wedge F \wedge (F \leftrightarrow E)$

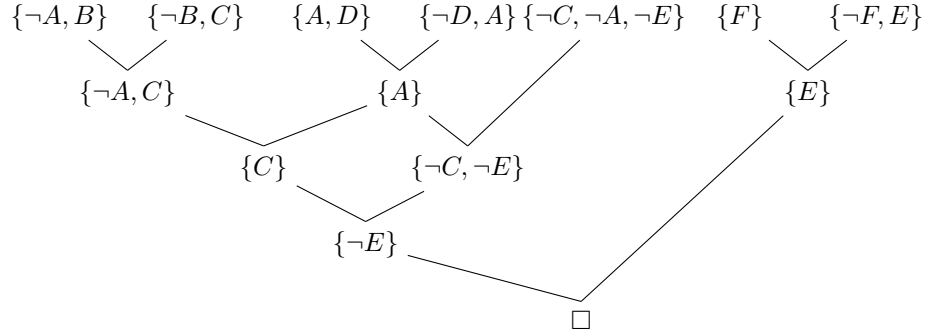
Solution:

(a) $\Delta_1 = \{\{A, B, C\}, \{\neg A, B, C\}, \{A, \neg B, C\}, \{A, B, \neg C\}, \{\neg A, \neg B, C\}, \{\neg A, B, \neg C\}, \{A, \neg B, \neg C\}, \{\neg A, \neg B, \neg C\}\}$



(b) $\Delta_2 = \{\{G, \neg H\}, \{\neg G, H\}, \{\neg A, B\}, \{\neg B, C\}, \{A, D\}, \{\neg C, \neg A, \neg E\}, \{\neg D, A\}, \{F\}, \{\neg F, E\}, \{\neg E, F\}\}$

Resolution tree:



Exercise 3

Transform the following formulas to CNF. To do so, follow the steps from the lecture (Chapter 7 slide 26) and give the intermediate results (you may skip those steps where there is nothing to do for the formula at hand). Simplify the resulting formulas where possible.

- (a) $(P \rightarrow Q) \leftrightarrow (P \rightarrow R)$
- (b) $\neg(P \leftrightarrow Q) \vee (Q \rightarrow R)$
- (c) $(P \rightarrow R) \wedge (Q \rightarrow R) \wedge \neg(\neg Q \wedge (\neg R \vee P))$

Solution:

- (a) 1. Eliminate $\leftrightarrow$: $((P \rightarrow Q) \rightarrow (P \rightarrow R)) \wedge ((P \rightarrow R) \rightarrow (P \rightarrow Q))$
 2. Eliminate $\rightarrow$: $(\neg(\neg P \vee Q) \vee (\neg P \vee R)) \wedge (\neg(\neg P \vee R) \vee (\neg P \vee Q))$
 3. Move $\neg$ inwards: $((P \wedge \neg Q) \vee \neg P \vee R) \wedge ((P \wedge \neg R) \vee \neg P \vee Q)$
 4. Distribute $\vee$ over $\wedge$: $(\neg P \vee \neg Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$
 CNF: $(\neg P \vee \neg Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$
- (b) 1. Eliminate $\leftrightarrow$: $\neg((P \rightarrow Q) \wedge (Q \rightarrow P)) \vee (Q \rightarrow R)$
 2. Eliminate $\rightarrow$: $\neg((\neg P \vee Q) \wedge (\neg Q \vee P)) \vee \neg Q \vee R$
 3. Move $\neg$ inwards: $(P \wedge \neg Q) \vee (Q \wedge \neg P) \vee \neg Q \vee R$
 4. Distribute $\vee$ over $\wedge$: $\neg P \vee \neg Q \vee R$
 CNF: $\neg P \vee \neg Q \vee R$
- (c) 1. Elimination of equivalence: nothing to do
 2. Elimination of implication: $(\neg P \vee R) \wedge (\neg Q \vee R) \wedge \neg(\neg Q \wedge (\neg R \vee P))$
 3. Move negation inwards: $(\neg P \vee R) \wedge (\neg Q \vee R) \wedge (Q \vee (R \wedge \neg P))$
 4. Distribute $\vee$ over $\wedge$: $(\neg P \vee R) \wedge (\neg Q \vee R) \wedge (Q \vee R) \wedge (Q \vee \neg P)$
 CNF: $(\neg P \vee R) \wedge (\neg Q \vee R) \wedge (Q \vee R) \wedge (Q \vee \neg P)$

Exercise 4

Consider the formulas in Exercise 3. Are they satisfiable? Draw the truth table and give the number of interpretations that satisfy the formula.

Solution:

$$\gamma = (\neg P \vee \neg Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$$

P	Q	R	$(\neg P \vee \neg Q \vee R)$	$(\neg P \vee Q \vee \neg R)$	γ
$\perp$	$\perp$	$\perp$	$\top$	$\top$	$\top$
$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$
$\perp$	$\top$	$\perp$	$\top$	$\top$	$\top$
$\perp$	$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\top$	$\top$	$\perp$	$\perp$
$\top$	$\top$	$\perp$	$\perp$	$\top$	$\perp$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$

$$\gamma = \neg P \vee \neg Q \vee R$$

P	Q	R	γ
$\perp$	$\perp$	$\perp$	$\top$
$\perp$	$\perp$	$\top$	$\top$
$\perp$	$\top$	$\perp$	$\top$
$\perp$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\top$
$\top$	$\perp$	$\top$	$\top$
$\top$	$\top$	$\perp$	$\perp$
$\top$	$\top$	$\top$	$\top$

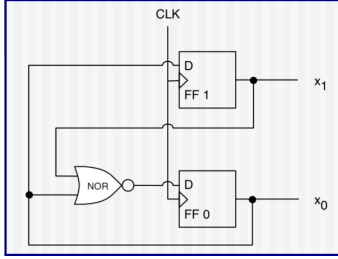
$$\gamma = (\neg P \vee R) \wedge (\neg Q \vee R) \wedge (Q \vee R) \wedge (Q \vee \neg P)$$

P	Q	R	$(\neg P \vee R)$	$(\neg Q \vee R)$	$(Q \vee R)$	$(Q \vee \neg P)$	γ
$\perp$	$\perp$	$\perp$	$\top$	$\top$	$\perp$	$\top$	$\perp$
$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\perp$	$\top$	$\perp$	$\top$	$\perp$	$\top$	$\top$	$\perp$
$\perp$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\perp$	$\top$	$\perp$	$\perp$	$\perp$
$\top$	$\perp$	$\top$	$\top$	$\top$	$\top$	$\perp$	$\perp$
$\top$	$\top$	$\perp$	$\perp$	$\perp$	$\top$	$\top$	$\perp$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$

The formula is satisfiable and has 3 satisfying assignments.

Exercise 5

Encode the example in the slides of Chapter 7 as a logical formula to verify that if the counter is initialized in the range 0-2, it cannot transition to the value 3.



- Counter, repeatedly from $c = 0$ to $c = 2$.
- 2 bits x_1 and x_0 ; $c = 2 * x_1 + x_0$.
- (“FF” Flip-Flop, “D” Data IN, “CLK” Clock)

→ The circuit simply repeats the operations:
 $x_0 \leftarrow NOR(x_0, x_1) = \neg(x_0 \vee x_1)$; and $x_1 \leftarrow x_0$. The clock is there so that both operations happen simultaneously, so x_0 and x_1 are updated at the same time.

Hint: you need to refer to two different states of the counter (before and after an operation), so you’ll need different propositions for those.

Hint: If you want to prove that given your knowledge encoded as a formula ϕ_K , some statement ϕ_S is true, then you need to encode the formula $\phi = \phi_K \wedge \neg\phi_S$. If ϕ is unsatisfiable, then it means that if we assume ϕ_K , ϕ_S must be true (as $\neg\phi_S$ leads to contradiction).

Solution:

We encode the formula in term of propositions x_0, x_1, x'_0, x'_1 , which encode the state of the circuit before and after an operation.

We know that:

- The value of the circuit is initially in the range 0-2: $\neg(x_0 \wedge x_1)$
- The value of the circuit is updated with $x_0 \leftarrow \neg(x_0 \vee x_1)$: $x'_0 \leftrightarrow \neg(x_0 \vee x_1)$
- The value of the circuit is updated with $x_1 \leftarrow x_0$: $x'_1 \leftrightarrow x_0$

We want to prove that $\neg(x'_0 \wedge x'_1)$. So, our full formula is: $(\neg(x_0 \wedge x_1)) \wedge (x'_0 \leftrightarrow \neg(x_0 \vee x_1)) \wedge (x'_1 \leftrightarrow x_0) \wedge (x'_0 \wedge x'_1)$

Note: The encoding above is just one possibility. For example, one can also use propositions $c_0, \dots, c_3, c'_0, \dots, c'_3$ representing the value of the counter, and add clauses making sure they have the right values such as $c_0 \leftrightarrow \neg x_0 \wedge \neg x_1$. This however, would not scale if the counter can have millions of values